Sweet Spot:
Choosing the Price and Quantity That Maximize Profit

ECON 201: Principles of Microeconomics
Lecture 9

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Boba Break

Boba Break is a stall at the Tuesday farmers market on campus. What price should it charge for a cup of boba to maximize its profit?

$$\text{Profit} = \underbrace{P \times Q}_{\text{revenue, } R} - \underbrace{C(Q)}_{\text{cost}}$$

- In choosing the price, they need to consider their costs, given by the **cost function** $C(Q)$.
- Boba Break also faces the tradeoff: a higher price P means more revenue per cup, but it will sell fewer cups Q (**demand curve**).

Boba Break's Costs

Boba Break pays $100 for the stall per day and it costs $4 to make each cup of boba. So the total cost is

$$C(Q) = 100 + 4Q$$

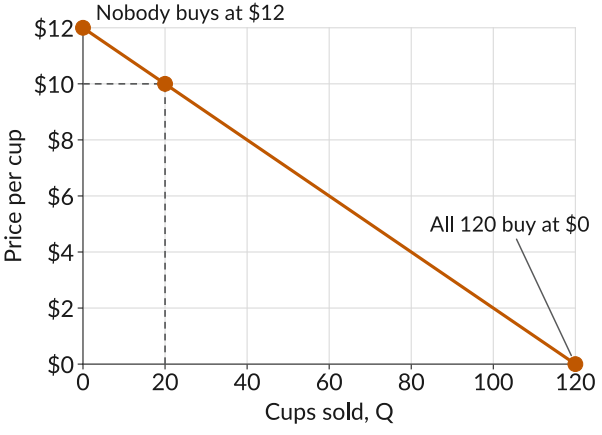
- Remember that the **average cost** is the cost per unit, $\text{AC} = C(Q)/Q$
- The **marginal cost** is the increase in total cost for every additional unit, $\text{MC} = \Delta C/\Delta Q$

Q	$C(Q)$	AC	MC
20	180	9.00	4
30	220	7.33	4
40	260	6.50	4
50	300	6.00	4
60	340	5.67	4

Boba Break's Demand Curve

- About 120 students walk past the stall.
- Their willingness to pay is spread evenly from $0 to $12, so every $1 off the price sells 10 more cups.
- So the demand curve is:

$$P = 12 - \frac{Q}{10}$$



Revenue, Cost, and Profit at 20 Cups

How do we get Boba Break's profit for any quantity Q?

- To sell $Q = 20$ cups, the demand curve says the price has to be $P = 12 - Q/10 = 12 - 20/10 = 10$.
- Revenue is $R = P \times Q = 10 \times 20 = 200$.
- Cost is $C(Q) = 100 + 4Q = 100 + 4 \times 20 = 180$.
- Profit is $R - C(Q) = 200 - 180 = 20$ for the day.

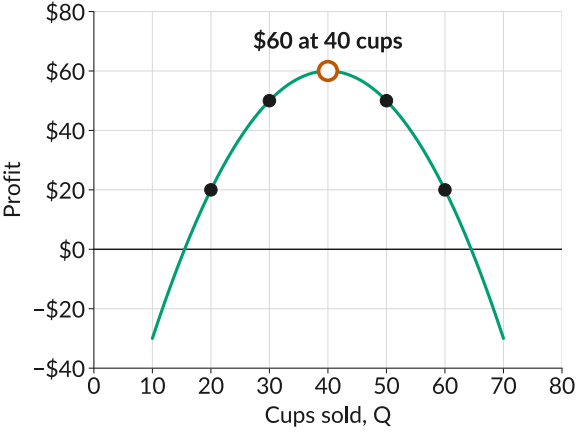
Revenue, Costs, and Profit

Doing the same for several quantities gives the table below.

Q	P	R	$C(Q)$	Profit
20	10	200	180	20
30	9	270	220	50
40	8	320	260	60
50	7	350	300	50
60	6	360	340	20

Profit at Each Quantity

- Profit rises to 40 cups and falls after it.
- At 40 cups the price is $8 and profit is $60 for the day.
- Selling more cups means cutting the price on **every** cup, and past 40 that costs more than it brings in.



Marginal Revenue

The **marginal revenue** is the increase in revenue when one additional unit of output is sold:

$$\text{MR} = \frac{\Delta R}{\Delta Q}$$

Example: Boba Break goes from 20 cups to 21.

- To sell 20 cups, the price has to be \$10, so revenue is $10 \times 20 = 200$.
- For 21 cups it has to be \$9.90, so revenue is $9.90 \times 21 = 207.90$.
- Revenue goes from \$200 to \$207.90, so MR $= 207.90 - 200 = 7.90$.
- Marginal cost is constant here: going from 20 cups to 21, cost rises by \$4.

Marginal Revenue at Several Quantities

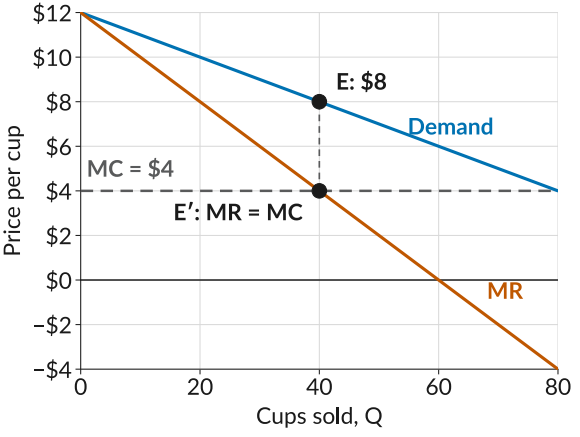
Here is the marginal revenue from one more cup at each quantity. Where is it above the marginal cost of $4, and where is it below?

Q	$R(Q)$	$P(Q+1)$	$R(Q+1)$	MR	MC
20	200	9.90	207.90	7.90	4
30	270	8.90	275.90	5.90	4
40	320	7.90	323.90	3.90	4
50	350	6.90	351.90	1.90	4

Note: marginal revenue is always less than the price. To sell one more cup, Boba Break has to lower the price on every cup it was already selling: at 20 cups the 21st cup sells for $9.90, but revenue rises by only $7.90.

Marginal Revenue Meets Marginal Cost

- Marginal revenue at every quantity gives the **marginal revenue curve**. It lies below the demand curve, because MR is less than the price.
- Marginal cost is flat at $4.
- The two cross at 40 cups, at *E′*. Going up to the demand curve at *E* gives the price, $8.



The Rule for Choosing Quantity

- While marginal revenue is above marginal cost, selling one more cup adds to profit.
- Once marginal revenue falls below marginal cost, the extra cup costs more than it earns.
- The 40th cup adds $4.10 to revenue and the 41st only $3.90, against a cost of $4 each.
- The **profit-maximizing quantity** is where the two are equal, MR = MC: for Boba Break, 40 cups, sold at $8 a cup.

Does the Stall Fee Change the Best Quantity?

- At each step, Boba Break asks: should it sell one more cup? The answer depends only on what that cup adds to revenue (MR) and to cost (MC).
- The $100 stall fee is the same whether Boba Break sells 39 cups or 41, so it never enters that comparison.
- Check: raise the fee to $150, and profit at 20, 30, 40, 50, and 60 cups goes from 20, 50, 60, 50, 20 to −30, 0, 10, 0, −30. It still peaks at 40 cups.
- Boba Break makes the same choice of price and quantity, but earns $50 less.
- The fee does matter for a different decision: whether to set up the stall at all. At 40 cups, revenue minus the cost of the cups is $320 − $160 = $160, so the stall is worth it while the fee is below $160.

What to Do Next

- Before next class: read section **7.6**, and go over the **slides** and the **worksheet**.
- Section 7.6 also finds the best price and quantity using **isoprofit curves**. We skip those and use marginal revenue and marginal cost, as we did today.
- Work through the **practice problems** for this lecture.
- Next class we ask who gains from the sale, the buyer or the seller, and by how much.
- **Quiz 4 is Monday, September 28**, and covers Lectures 8 and 9.